Rewarding Learning

ADVANCED SUBSIDIARY (AS)
General Certificate of Education
2017

Centre Number

Candidate Number

Digital Technology

Assessment Unit AS 1

assessing

Approaches to System Development

[CODE]

SPECIMEN PAPER

TIME

1 hour 30 minutes.

INSTRUCTIONS TO CANDIDATES

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
Write your answers in the spaces provided in this question paper.
Answer **all five** questions.

INFORMATION FOR CANDIDATES

The total mark for this paper is 100.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each question or part question.
Quality of written communication will be assessed in questions **2(e)**, **4(d)** and **5(d)**.

For Examiner's use only	
Question Number	**Marks**
1	
2	
3	
4	
5	
Total Marks	

Version 2: 6 March 2017

Examiner Only	
Marks	Re-mark

1 The 'software crisis' has changed the way in which large information systems are developed.

(a) Explain the main impact of the 'software crisis'.

_____

_____

_____

_____

_____

_____

_____

_____ [4]

(b) Identify **two** specific duties of the project manager during system development.

1 _____

_____

2 _____

_____ [2]

Fact-finding can be used to identify the user requirements during the analysis stage.

Examiner Only	
Marks	Re-mark

(c) Describe how document sampling can be used during fact finding.

[3]

(d) Using examples from a stock control system, illustrate the difference between functional and non-functional user requirements.

[4]

Customers of a bookshop can order a book by telephone or email. The book is posted to the customer. At the end of the month, the customer receives an invoice. Customers can make payments by telephone or post, after which they receive a receipt by post.

(e) Produce a context data flow diagram for this book ordering system.

[6]

Examiner Only	
Marks	Re-mark

2 During the design stage of system development the test plan is produced.

Examiner Only	
Marks	Re-mark

(a) Identify **three** other activities you would expect to take place during system design.

1 _____

_____

2 _____

_____

3 _____

_____ [3]

(b) Describe how storyboarding can be used during the development of a web-based application.

_____

_____

_____

_____

_____

_____

_____

_____ [4]

(c) Identify **two** main components of a test plan.

1 _____

_____

2 _____

_____ [2]

During the implementation stage the new system replaces the existing system.

(d) Explain what happens during data conversion.

_____

_____

_____

_____

_____

_____ [3]

Examiner Only	
Marks	Re-mark

An organisation decides to use direct changeover for its new payroll system.

(e) Evaluate the organisation's choice of changeover method.

Quality of written communication will be assessed in this question

[9]

Examiner Only	
Marks	Re-mark

Examiner Only	
Marks	Re-mark

3 An information system is being developed using the waterfall model.

(a) Describe the main features of the waterfall model.

[4]

The project manager identifies the project's constraints and risks.

(b) Identify **two** risks to a typical software project and explain how each risk could affect the project.

1

2

[4]

The project manager uses Gantt charts and critical path analysis (CPA).

Examiner Only	
Marks	Re-mark

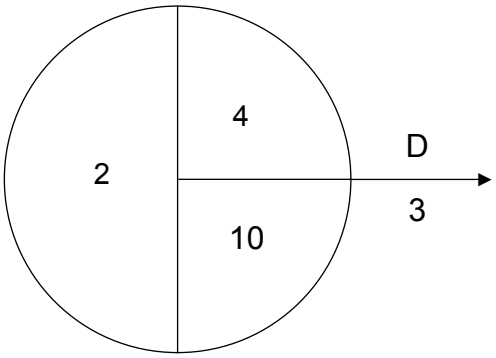
(c) Identify the main features of a Gantt chart.

[4]

(d) Describe the use of CPA in project management.

[4]

(e) Examine the following node from a CPA diagram:



Complete the following table:

Label	Description
2	The node number
4	
10	
D	
3	

[4]

Examiner Only	
Marks	Re-mark

Examiner Only	
Marks	Re-mark

4 After the implementation stage is complete, information systems still require maintenance.

(a) Describe what is meant by corrective maintenance.

[2]

(b) Explain why maintenance is required even when a system has been fully tested during development.

[6]

(c) Describe the contents of a disaster recovery plan.

Examiner Only	
Marks	Re-mark

[4]

A particular organisation decides to use a combination of full backups and differential backups.

Examiner Only	
Marks	Re-mark

(d) Evaluate the organisation's decision.

Quality of written communication will be assessed in this question

[9]

Examiner Only	
Marks	Re-mark

5 A programmer uses an Integrated Development Environment (IDE) during the development of new software.

(a) List **two** facilities provided by an IDE.

1 _____

_____

2 _____

_____ [2]

The programmer uses iteration to solve a problem.

(b) Explain how iteration can be used to perform a particular calculation one hundred times.

_____

_____

_____

_____

_____

_____

_____

_____ [4]

Modern programming languages support object-orientated programming.

Examiner Only	
Marks	Re-mark

(c) What is meant by each of the following terms?

Object

Inheritance

[4]

(d) Evaluate the use of the object-oriented approach in improving the quality of the software produced.

Quality of written communication will be assessed in this question

[9]

Examiner Only	
Marks	Re-mark